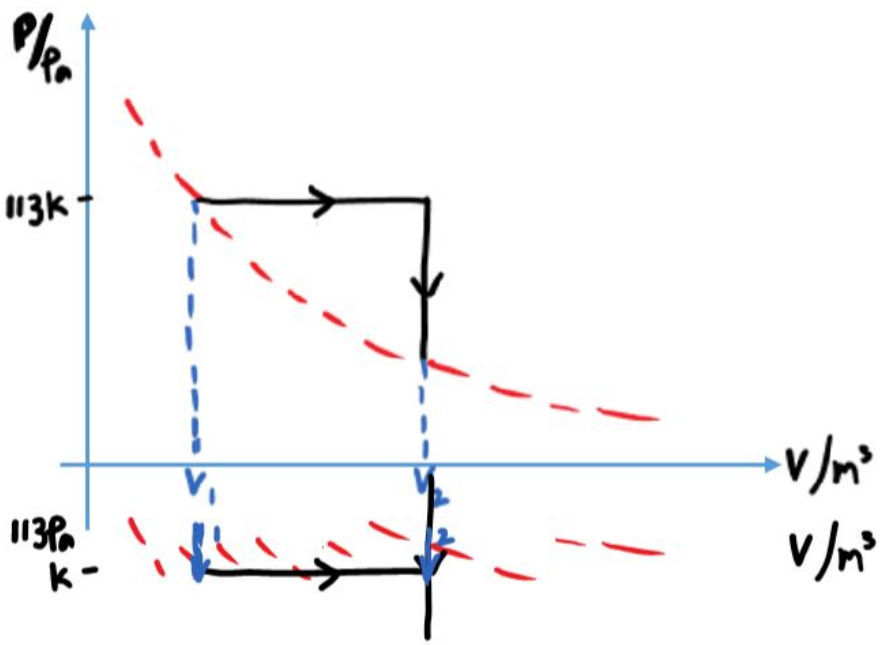
Qn			Answers	Marks
7	(a)	(i)	The **specific heat capacity**, *c*, of a body is defined as the amount of thermal energy required to raise the temperature of unit mass of the substance by one degree. The S.I. unit is J kg^{-1} K^{-1}.	B1
	(a)	(ii)	The same steady inflow and outflow temperatures are maintained for both experiments so that the rate of heat lost is the same in both experiments so that it can be taken into account in the conservation of energy equations when calculating the specific heat capacity.	B1
	(a)	(iii)	$V_1 I_1 = \frac{m_1}{t_1} c\Delta\theta + \frac{H_1}{t_1}$ ---------- (1) $V_2 I_2 = \frac{m_2}{t_2} c\Delta\theta + \frac{H_2}{t_2}$ ---------- (2) To award one mark if students managed to relate power to rate of energy absorption, but missed out on heat loss.	M1
			Since $\frac{H_1}{t_1} = \frac{H_2}{t_2}$, $c = \frac{(V_2 I_2 - V_1 I_1)}{(\frac{m_2}{t_2} - \frac{m_1}{t_1})\Delta\theta} = \frac{(37.8 - 25.2)(60)}{(0.232 - 0.150)(17.4 - 15.2)}$	M1
			= 4190 J kg^{-1} K^{-1}	A1
	(b)	(i)	1. Isobaric expansion / expansion under constant pressure	B1
			2. Isovolumetric cooling / cooling under constant volume	B1

	(b)	**(ii)**	$F = mg = 8.00\ (9.81) = 78.5$ N	M1
			$P = 78.5/(60.0 \times 10^{-4)} + 100000$ Pa	C1
			$= 1.13 \times 10^5$ Pa	A1
	(b)	**(iii)**	 (Not necessary to draw in the isotherm.)	
			-1 mark for expansion under constant pressure (horizontal to the right)	B1
			-1 mark for cooling under constant volume (vertical line downwards)	B1
			Lose second mark if line is disjointed, or not in correct order.	
	(b)	**(iv)**	For heating, $\Delta U = Q_1 + W = Q_1 + p\Delta V$ For cooling, $W = 0\ \therefore \Delta U = Q_2$	B1
			Since ΔU is the same for both processes, $Q_1 + p\Delta V = Q_2$ Hence net heat gained by gas $= Q_1 - Q_2$ $= p\Delta V$ $= 113000 \times 0.200 \times 60.0 \times 10^{-4}$ $= 136$ J OR $\Delta U_{net} = Q_{net} + W_{net}$ Since there is no change in the temperature between (a) and (c), $\Delta U_{net} = 0$ $Q_{net} = -W_{net}$ $= 1.13 \times 10^5 \times 0.200 \times 60.0 \times 10^{-4}$ $= 136$ J	M1 M1 A1 B1 M1 M1 A1

	(c)	(i)	As the air expands, its pressure decreases. The lower air pressure produces a smaller downward force on the water.	B1
			According to Newton's third law of motion, the water exerts an equal and opposite smaller force on the bottle, causing it to move upwards.	B1
	(c)	(ii)	Select two suitable points from the graph, for example: $p_1 \approx 3.6 \times 10^5$ Pa, $V_1 \approx 1.1 \times 10^{-3}$ m^3 $p_2 \approx 1.0 \times 10^5$ Pa, $V_2 \approx 3.0 \times 10^{-3}$ m^3 $p_1V_1 \approx 4.0 \times 10^2$ J $p_2V_2 \approx 3.0 \times 10^2$ J.	M1
			Therefore, pV **decreases** during the expansion. Since the amount of gas is constant and $pV = nRT$, the temperature of the air **decreases** as the air expands.	A1